

01 · Learning corrections without losing the rest

Active Inference in the Extremes

Can a small adaptive memory improve a frozen language model while limiting the damage elsewhere?

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A working correction system, mixed predictive-coding results, and a useful bound on harm.

02 · The original bet: make learning selective

Attach a small learning system to a frozen transformer. Read its internal state, then add a bounded correction where it helps.

Preserve useful knowledge

Learn a supplied answer without needlessly changing unrelated predictions.

Share what should transfer

Recognize the same fact in different wording, while keeping nearby facts distinct.

Test the mechanism and cost

Separate representation, learning signal and retrieval. Measure useful retention, interference and compute together.

The first cap was motivated by predictive coding. Later experiments explicitly tested error inference.

03 · Three ideas guided the programme

We aimed to connect selective learning, decisions under uncertainty and the consequences of extreme errors.

Active inference

Update beliefs and choose actions using a model of observations, preferred outcomes and information value. Our autonomous action-selection loop remains future work.

Predictive coding

Use mismatches between predictions and observations to infer corrections. We tested iterative error inference in acquisition and reader training.

Heavy-tailed distributions

Large deviations can remain more probable than an exponential tail predicts. We measured harmful prediction changes and tested tail descriptions over the observed range.

The integration is incomplete. The experiments still reveal which components help, fail or cost too much.

04 · What the cap actually does

The main testbed uses GPT-2 small: 124 million parameters, with its base weights frozen.

A new question

The transformer computes internal features. The cap reads features at blocks 4, 8 and 12.



Memory and a reader

The reader compares the question with stored facts. A gate can select a correction or choose none.



A corrected prediction

Bounded residual vectors are added at the write sites. The transformer produces its next-token probabilities.

Frozen weights do not guarantee frozen behavior: changing an activation changes what happens downstream.

05 · Who supplies the answer to be learned?

The targets were fixed in downloaded datasets before our runs. Our experiment did not invent replacement answers on the fly.

zsRE: teach a reference answer

Prompt: “What league was Sporting Canamy?”

Target: Tercera División de México

We used the record's first reference answer, `answers[0]`. This is answer acquisition, not a deliberately false replacement.

CounterFact: teach a controlled replacement

Prompt: “James Howell speaks”

Stored original: English · New target: Spanish

We used `target_new` from the record. Success means following the supplied edit; Spanish is not being asserted as historical truth.

A dataset's original-answer label is separate from what our frozen model actually generated.

06 · Does the answer survive different wording?

Teach: “What league was Sporting Canamy?” → Tercera División de México

Test paraphrase: “What league did Sporting Canamy join with?”

After 1,000 edits	Original prompt	Paraphrase
Learned reader	Tercera División de México	Tercera División de México
Random reader	Tercera División de México	CD Atlético Baleares
Stable v0 cap	[empty answer]	forward

The alternative question came from zsRE's rephrase field. During testing, the model saw the question without the target answer.

Remembering the teaching prompt and recognizing the same request are different achievements.

07 · A paraphrase can also add distracting context

Teach: “James Howell speaks” → Spanish

Test: “Cataraqui is also the name of a municipal electoral district. The language used by James Howell is”

After 1,000 edits	Original prompt	Paraphrase
Learned reader	Spanish	Spanish
Random reader	Spanish	Ukrainian
Stable v0 cap	Spanish	"the city of the people."

The entire test prompt came from CounterFact's paraphrase_prompts. We retained its unrelated opening sentence.

We test the supplied alternative wordings. They are not a guarantee of understanding every possible question.

08 · An edit must pass more than one test

Generated answers stop at newline or end-of-sequence, with a 32-token limit. Editing scores use complete-answer matches to accepted variants.

— Acquire and retain

Can the cap produce the target now, and again after many later edits?

— Generalize the wording

Can it answer the supplied paraphrases? Retained paraphrase score is the main editing outcome.

— Preserve and revise

Do unrelated and nearby prompts retain the base's output? Can a newer answer replace an older one?

— Inspect probabilities

On ordinary text, compare the full token distribution and the probability of the actual next token.

Preservation means preserving behavior. If the base gave a bad answer, preserving it is still a pass.

09 · Recalling a fact is only the first step

The two supplied papers help locate our work within a larger question: can a system use changed knowledge consistently?

MQuAKE: consequences of an edit

The paper tests questions that require chains of facts after an edit. Our runs extracted single-fact changes from MQuAKE-CF and tested their alternative prompts at 300 edits.

CounterBench: causal counterfactuals

The paper asks what follows when variables in an explicit causal system are changed. It studies causal inference, rather than simply recalling an edited association.

We did not run CounterBench or MQuAKE's full multi-hop benchmark. Those remain stronger future tests.

10 · The central comparison changes the reader

Same main-study base and supplied edit streams. These three conditions make the retrieval question easiest to see.

— Learned reader

A trained reader routes questions to the correction memory and can abstain.

— Random reader

Random-geometry retrieval with a gate tests how much the learned representation contributes.

— Stable v0 cap

The earlier activation-memory cap provides a reference for the newer reader architecture.

Three subject realizations × five orders. zsRE and CounterFact reach 1,000 edits; MQuAKE reaches 300 and is descriptive.

Five orders reuse a realization's facts. They do not turn three subject populations into fifteen.

11 · Learning the reader improves paraphrase retention

Retained paraphrase score, averaged over orders within each realization, then over the three realizations. Higher is better.

Dataset and endpoint	Learned reader	Random reader	Stable v0 cap
zsRE · 1,000 edits	96.0%	52.4%	18.6%
CounterFact · 1,000 edits	67.8%	12.1%	0.0%
MQuAKE · 300 edits	71.6%	0.0%	0.0%

The learned-reader realization means span 95.5–96.5% on zsRE, 66.6–69.2% on CounterFact and 70.3–72.3% on MQuAKE.

The cap often carries an answer beyond its teaching prompt. The strength of that transfer depends on the dataset.

Source: Stage-4 triplet summary. Bars start at zero on a common 0–100% scale; ranges are descriptive.

12 · Successful edits can still disturb other predictions

We replayed ordinary text at identical prefixes, comparing capped and uncapped predictions at 245,237 positions per cell.

Distribution change: KL

KL compares the whole next-token probability distribution. All 45 main learned-reader cells exceeded the registered mean-KL limit of 0.001.

Observed-token harm: Δ

$\Delta = \log(p_{\text{base}} / p_{\text{cap}})$ for the token actually present in the text. Positive values mean that token became less likely. Negative values are improvements.

Why a small mean can mislead

Most positions change little. A small minority can suffer much larger losses. Measure how often, how severely and how far into the tail.

The runs passed data-integrity checks. The learned reader failed the specified distribution-fidelity benchmark.

13 · How often harm occurs and how large it is

Count a harmful change when Δ exceeds 0.01 nat. Severity is the mean Δ among those harmful positions.

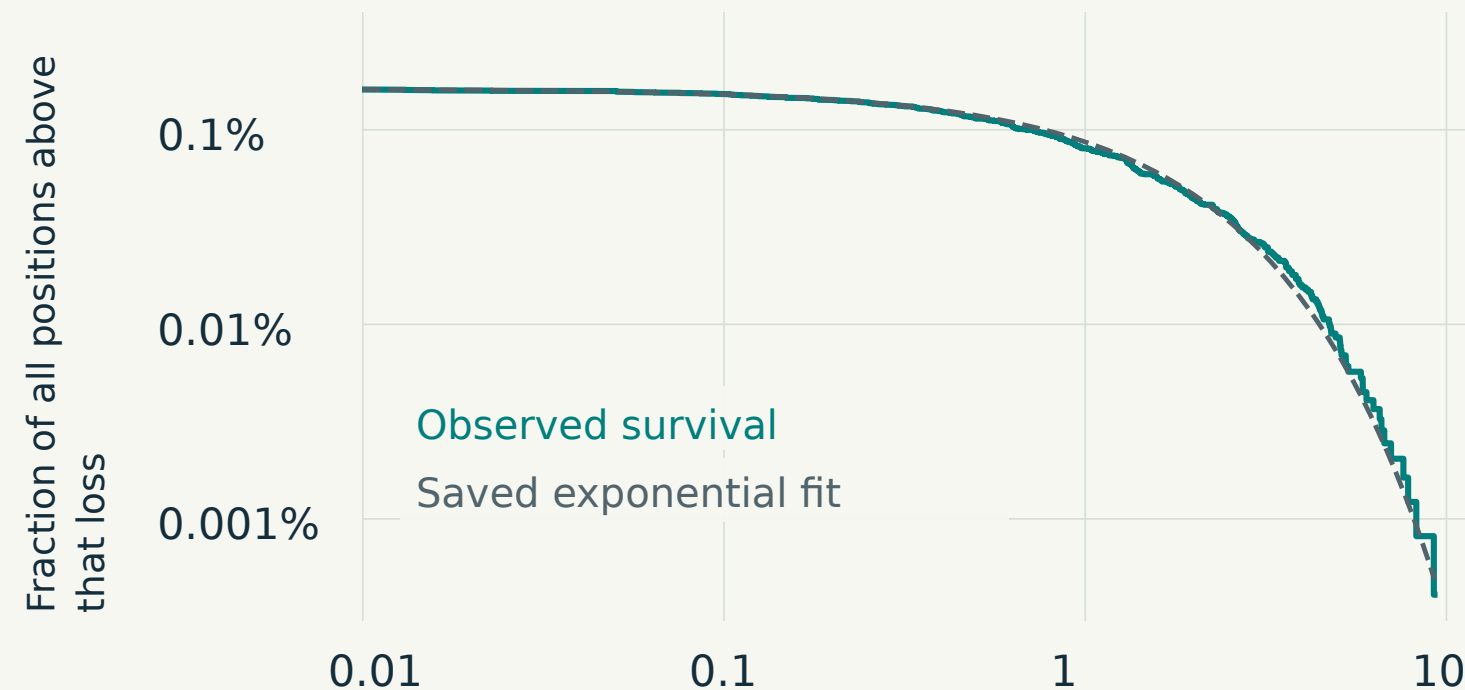
Dataset and cap	Harmful positions	Conditional severity
zsRE · learned	0.1594%	1.630 nats
zsRE · random	0.0266%	1.806 nats
zsRE · stable	0.1041%	1.716 nats
CounterFact · learned	0.2984%	1.913 nats
CounterFact · random	2.0450%	3.486 nats
CounterFact · stable	0.0000%	Undefined

Zero observed CounterFact harm for stable v0 accompanies zero paraphrase retention. Quietness alone is insufficient.

14 · Extreme losses have different observed tails

A survival curve shows the fraction of positions whose loss increase exceeds a chosen size. Farther right means a larger loss.

zsRE · learned reader



zsRE · stable v0



Stable v0 has a more pronounced tail here. The learned reader gains little from a more flexible tail fit.

15 · A probability floor gives a bound we can prove

Mix the original next-token distribution with the cap's distribution at the same text prefix.

Keep some of the base

$$\rho = \exp(-1) \approx 0.368$$

About 37% of the original distribution remains in the mixture.



Add the corrected distribution

$$q = \rho p_{\text{base}} + (1 - \rho) p_{\text{cap}}$$

About 63% comes from the capped model.



Bound the loss increase

$$q \geq \rho p_{\text{base}}$$

$$\log(p_{\text{base}} / q) \leq 1 \text{ nat}$$

This bounds a token's loss at a shared prefix. It is not a one-nat bound on a whole generated answer.

16 · The mixture reduced severity to about one-third

Mean loss increase among positions with $\Delta > 0.01$ nat, on ten exposed memories: two datasets × five orders at 300 edits.

Dataset	Original cap	Probability mixture
zsRE	1.619 nats	0.542 nats
CounterFact	1.925 nats	0.581 nats

All ten memories met the declared retention-and-preservation rule. Paraphrase retention was unchanged on zsRE and fell by 0.67–1.17 percentage points on CounterFact.

Harm frequency barely changed. CounterFact mean KL still exceeded 0.001.

This is a measured reduction in the severity of unintended changes, with a small editing trade-off.

17 · “Using predictive coding” has several meanings

Changing the base representation, the acquisition signal and the reader's training rule are separate interventions.

Base representation

The legacy study uses a frozen 50M ePC base. The main study uses frozen 124M GPT-2. Results across these models are not a controlled representation comparison.

Acquisition credit

Hold the base and reader fixed. Change the signal that tells a new memory correction which direction to move.

Reader training

Hold the base fixed. Train the retrieval/controller network with backpropagation or iterative error inference, then acquire fresh memories.

The next results isolate acquisition credit and reader training within their own paired experiments.

18 · What our error-inference experiment computes

Both acquisition arms start with the same target and use the same bounded memory-update rule.

Adjoint credit

Differentiate the answer loss through the frozen model. Use the negative, normalized gradient to direct the correction.

Error-inference credit

Hold existing writes fixed. Start temporary site errors at zero. Minimize a quadratic error penalty plus the answer loss, then normalize the inferred errors.

$$E(e) = \frac{1}{2} \sum ||e||^2 + \text{answer loss with } e$$

The default uses eight settling steps at rate 0.1. A one-step normalized update matches the adjoint direction to numerical tolerance.

This implementation uses automatic differentiation. Its PC component is iterative error inference, not biological locality.

19 · More settling bought retention at a cost

Legacy 50M ePC base with the v0 live cap: zsRE, three exposed realizations, order 100, after 1,000 edits.

Settling steps	Original-prompt retention	Paraphrase retention	Learning time per cell
1	51.5%	13.1%	5.7 min
8	53.4%	13.1%	10.2 min
32	56.5%	12.2%	25.4 min

Mean ordinary-text loss increase rose from 0.00145 to 0.00221 nats between one and 32 steps. CounterFact paraphrase retention stayed at zero.

More inference preserved more teaching prompts, but did not establish better transfer or lower harm.

Source: PC-controls report. Means over three realizations; timing is learning only, not whole-process time.

20 · With the learned reader, credit effects were small

Frozen GPT-2 and the selected v5 reader; only acquisition credit changes. Each arm acquires 300 edits.

Dataset	Credit	Paraphrase retention	Acquisition process	Largest token loss
zsRE	Adjoint	98.3%	316 s	9.26 nats
zsRE	Error inference	98.3%	426 s	12.27 nats
CounterFact	Adjoint	80.7%	297 s	11.74 nats
CounterFact	Error inference	80.3%	359 s	12.23 nats

Mean harm improved on zsRE and worsened on CounterFact. Small score differences are not proof of equivalent behavior.

This transfer check provides no clear practical advantage for eight-step error credit.

21 · Training the reader with PC gave mixed outcomes

Three paired training seeds, the same 300 updates and 3.35M reader parameters. All evaluations use adjoint acquisition and 300 edits.

Dataset and training seed	BP paraphrase retention	ePC paraphrase retention
zsRE · seed 0	97.3%	94.7%
zsRE · seed 1	99.0%	97.0%
zsRE · seed 2	97.7%	98.0%
CounterFact · seed 0	81.0%	56.8%
CounterFact · seed 1	80.0%	77.8%
CounterFact · seed 2	83.5%	66.0%

CounterFact paraphrase retention fell in every pair. Its ordinary-text harm changed direction across seeds.

22 · The reader-training cost difference was large

Whole-training process time for the same 300 updates, including construction and caching. Each pair starts from matched parameters.

Training seed	Backpropagation	Error-inference training
Seed 0	0.241 h	24.579 h
Seed 1	0.252 h	24.112 h
Seed 2	0.242 h	23.432 h

Measured paired cost ratios: 102.1×, 95.8× and 96.7×. Backpropagation took about 14–15 minutes per reader; ePC took about a day.

The installed ePC training procedure was substantially more expensive, without a consistent efficacy-and-harm gain.

Source: Completed PC-reader process receipts. Acquisition and evaluation are separate; bars share a zero origin and scale.

23 · Last-layer writes increased measured harm

Last-only versus all-site writes, with readers using either all taps or the upper two. Bars show the ratio of mean signed loss increases.

Dataset and training seed	All read taps	Upper read taps
zsRE · seed 0	3.82×	2.64×
zsRE · seed 1	3.38×	2.71×
zsRE · seed 2	3.17×	4.37×
CounterFact · seed 0	3.29×	2.97×
CounterFact · seed 1	2.87×	3.30×
CounterFact · seed 2	3.14×	3.63×

Upper-only reading also lowered CounterFact paraphrase retention in all three seeds.

Every write restriction raised this harm measure, while paraphrase scores changed by at most 0.67 percentage points.

24 · What the experiments changed in our thinking

Useful retrieval, a good learning direction and control of collateral effects must be evaluated separately.

Retrieval is central

Learning the reader greatly improves paraphrase retention. Restricting the interface to upper layers did not provide the hoped-for benefit.

PC still has trade-offs

An adjoint control used 53–58% of its offered budget. Credit direction versus actual compute remains unresolved.

A simple bound was useful

Mixing with the base reduced harmful-event severity while largely retaining edits. It did not solve every fidelity failure.

The wider theory is still open

The bounded κ -loss pilot missed its success rule. That pilot did not implement the full coupled-entropy/free-energy proposal.

The best next experiment should isolate one proposed mechanism and preserve both usefulness and harm as outcomes.

25 · After the conference: make the blanket measurable

A Markov blanket is a set of variables that makes internal and external states conditionally independent. Drawing a software boundary does not establish this property.

Specify the states

Name the environment, the adaptive cap's internal state and the frozen transformer's state in a probabilistic model.



Specify the interfaces

Define which observations enter, which corrections leave and what a proposed coupled or porous boundary actually asserts.



Test the asserted property

Use a small controlled model with known dependencies. Measure residual dependence and perturb the interface before scaling up.

The existing read/write interfaces are a starting point. A classical or coupled blanket requires its own stated test.

26 · Coupled entropy needs a complete objective

The proposed framework changes how probabilities and rare states contribute to uncertainty. Replacing one loss function is only part of that specification.

Define the distribution

Choose the modeled variable: correction state, residual or predicted token. Specify its support, normalization, units and reference distribution.

Define the mathematics

Agree the coupled logarithm, normalized weighting, scale/root and constraints. Include their derivatives when optimizing the chosen free energy.

Connect theory to a test

Recover the ordinary limit and check values and gradients against independent calculations. Then ask whether useful corrections cause less harm.

A fitted tail shape is not automatically the objective's coupling parameter. Our earlier κ pilot did not test this full model.

27 · First isolate the objective, then the learning rule

Proposed first comparison: an agreed ordinary objective versus its coupled counterpart, using the same architecture and learning method.

Pair the training

Three matched initializations, the same data and update schedule. Keep exact-gradient learning in both arms first, to isolate the objective's effect.

Measure the trade-off

Evaluate both readers on the same two 300-edit streams: 12 cells. Report paraphrase retention, preservation, harm frequency, severity, extremes and actual cost.

Extend only after that test

Add a matched predictive-coding comparison, then multi-hop or explicit causal tasks. Reused subjects support exploratory follow-up, not fresh confirmation.

The question is whether the coupled objective improves the usefulness-harm trade-off, including when the answer is negative.

28 · Toward an agent that chooses what to investigate

Active inference adds a decision: which observation or action is worth its cost, given current uncertainty and preferred outcomes?

Observe

A correction arrives, a paraphrase fails or a nearby fact changes unexpectedly.



Infer and choose

Model the uncertainty. Compare an audit, a correction and abstention by their expected consequences and information value.



Learn and test

Use the result to update beliefs. Compare the policy with fixed or random audits at the same resource budget.

For the conference talk: which result best explains the promise, and which limitation most needs a clearer explanation?